

# Strain Scores: Your Heart Rate Doesn't Know You Did Squats

A Strain score should answer the simple question: “how much load did my body take on today?”. This is different from readiness, as readiness asks whether the body looks prepared for strain; strain asks how much stress was actually accumulated. The score should therefore be driven by the work that happened and the body's response to it, with separate consideration for cardiovascular and muscular load. The mainstream wearable companies mostly agree with this, but they expose the logic differently.

WHOOP has made Strain one of its main scores. The publicly given explanation is centred mainly around cardiovascular strain, especially how long the user spends at elevated heart rates. This makes sense for cardio-heavy activity, however WHOOP has more recently added muscular load because pure heart-rate strain tends to undercount strength training. A heavy leg session can create real fatigue even if heart rate is not elevated for long enough to look like a hard run. [1]

Apple's Training Load is also useful because it separates today's workout effort from the longer-term interpretation of whether load is ramping up. Apple uses a 1-10 effort rating after workouts and automatically estimates effort for many cardio workouts using data like heart rate, GPS, elevation, age, height and weight, then compares the most recent 7 days with the previous 28 days. This is probably the cleanest mainstream version of the idea: first estimate how hard the activity was, then compare recent load to what the person normally does. [2]

Garmin has a wider training ecosystem, including acute load, load focus, recovery time and Training Readiness. Garmin's most useful contribution is that load is not just “calories burned” or “minutes exercised”. It is interpreted in the context of intensity, recovery and whether the user is building or overdoing training. [3]

Google's Cardio Load paper defines cardiovascular work accumulated across the day through heart-rate reserve (HRR), meaning the user's heart rate relative to the space between resting and maximum heart rate. This is a strong basis for the cardio part of strain because HRR is more personal than raw heart rate. A heart rate of 150 bpm may be fairly comfortable for one person and close to maximal for another. Google also uses an exponential curve, so harder minutes become disproportionately more valuable rather than treating every beat above rest in the same way. [4]

Moving to the scientific literature, the main history here starts with training impulse models like TRIMP. These models summarise training load from intensity and duration, usually using heart rate. The basic idea is still correct: the longer a person works at a high intensity, the more cardiovascular load they accumulate. Foster's session RPE work adds another useful layer by showing that perceived effort can be a practical way to quantify training load, especially when the available sensors miss some of the session. [5] [6]

The main scientific warning is that strain is good at measuring exposure and much worse at perfectly predicting injury or recovery. Acute workload ratio models compare recent load with what the athlete is used to, but review papers have shown problems with treating the ratio as a

magic injury predictor. The result can be statistically unstable, sensitive to the exact calculation and confounded by plenty of things that are not measured. Recent versus chronic load should therefore remain an interpretation tool rather than some sort of risk number. [7]

The strain calculation should treat cardiovascular and muscular load as its two additive channels. Distance, pace, power and elevation are evidence about the work, workout type tells the algorithm how to route that evidence, and RPE helps interpret missing effort. These inputs should support or modify the relevant channel rather than receive separate percentages, as doing so would often count the same session several times.

The clean structure would have 2 load channels:

- Cardiovascular load, calculated mainly from minute-level HRR.
- Muscular load, calculated from the exercises and sets that loaded the muscles.

External work, workout type and subjective effort will support those channels. They validate, replace or modify an estimate where appropriate, but they shouldn't receive independent percentages. This also agrees with the wider load monitoring literature: internal and external measures tend to be related, though the strength of that relationship changes substantially by training mode. They are useful accounts of the same work not interchangeable units. [8]

## Calculating cardiovascular load

Google's modified Banister calculation is a good starting point and can be used almost directly. For each minute:

$$\text{HRR} = \frac{\text{HR} - \text{RHR}}{\text{HR}_{\text{max}} - \text{RHR}}$$

$$\text{Cardio dose} = 0.64 \times \text{HRR} \times e^{k \times \text{HRR}}$$

HR is the time weighted average heart rate for that minute. RHR is the persons resting heart rate baseline and HRmax is their current credible maximum heart rate estimate. The coefficient k is 1.92 for males and 1.67 for females in Google's implementation. Where sex is unavailable, the midpoint of 1.795 is a reasonable neutral fallback. HRR should be clamped from 0 to 1 before the exponential is applied, mainly so that one optical sensor spike can't create a ridiculous amount of load. [4]

The minute then receives an eligibility multiplier:

- Below 30% HRR: 0
- From 30% to below 40% HRR:  $\frac{\text{HRR} - 0.30}{0.10}$
- At or above 40% HRR: 1

The linear 30-40% ramp is an implementation choice. Google states that load starts at 30% HRR and is down-weighted till 40%, but does not publish the exact function. A linear ramp is transparent, continuous and easy to replace if later validation suggests something better. The final minute value is Cardio dose × eligibility multiplier. Daily cardio load is then the sum of every eligible minute.

Incidental load also needs evidence of movement. This prevents a stressful meeting, caffeine or illness from being interpreted as exercise merely because heart rate rose. Minutes inside a manually logged or confidently auto-detected workout can pass the movement gate because the workout itself is evidence of activity. This follows Google's 24-hour extension of TRIMP and allows brisk walking, commuting and stairs to contribute without giving points for ordinary resting heart rate. [4]

The baseline values matter a lot. RHR should come from the stable estimate defined in the Personal Metric Baselines paper rather than whichever resting value happened to be recorded that morning. A recent measured HR<sub>max</sub> from a credible maximal effort or formal test is preferable to an age equation. When there is no usable observation,  $HR_{\max} = 208.9 - 0.74 \times \text{age}$  is a defensible population fallback from a large validation sample. Age equations still have large individual error, with one large study reporting a standard error of around 11 beats per minute, so an age-derived maximum should carry lower confidence and shouldn't be silently treated as a personal measurement. [9] [10]

The model should be conservative about automatically learning HR<sub>max</sub>. A new peak should only replace the estimate when it comes from sustained, high quality HR data during a credible hard workout. One isolated sample should be ignored. A measured peak that is slightly below an age estimate can still be useful evidence, but it should raise the personal estimate gradually unless the session was explicitly a maximal test. Personal Metric Baselines already describes the wider rules for baseline history, stale data and confidence.

Heart rate data should be handled minute by minute rather than reducing a workout to average HR. Intervals are the obvious reason: 20 minutes alternating hard and easy work should not look identical to 20 minutes at the same overall average. A minute is usable when at least 30 seconds contain valid heart rate samples. Gaps of no more than 90 seconds inside a recorded workout may be interpolated between good readings. Longer gaps remain missing. This is an engineering rule rather than a physiological threshold, and should be logged so it can be tested later.

External work is then used as a validator and fallback. Running pace, distance and gradient; cycling power; rowing pace or power; swimming distance; and similar measures can reveal that a session was real when wrist heart rate is poor. They should replace only the uncovered part of the workout. They should never be added on top of good HRR minutes as a second cardio score. Where a provider supplies reliable heart rate zones rather than samples, the midpoint HRR of each zone can be passed through the same equation for the time in that zone. Where neither heart rate nor zones exist, an endurance or team sport workout can use session RPE as a lower confidence fallback:

$$\text{Fallback cardio load} = \frac{\text{active minutes} \times \text{session RPE}}{10}$$

This keeps a 60-minute session rated 7/10 at 42 points rather than the 420 arbitrary units produced by raw session RPE. It is deliberately conservative and should only fill minutes with no better cardio estimate. A generic strength workout should not use overall session RPE to invent cardio load because that rating likely mostly describes local muscular effort. If a separate respiratory RPE is collected, it can use the fallback.

Calories and steps shouldn't be extra additive components. Both are useful for plausibility checks and missing data detection, but calories are partly model derived and steps are a poor description of cycling, rowing, swimming and lifting. When reliable HRR exists, adding either would mostly count the same work again.

## Calculating muscular load

As discussed in the Resistance Training and Muscular Load paper, muscular load is harder to quantify because it represents several overlapping types of stress rather than one directly measured quantity. That paper develops the set-dose model and explains the reasoning behind each input. Strain uses the resulting muscular dose through the following equations:

$$\text{Set dose} = \text{repetitions} \times \text{relative load} \times \text{exercise involvement} \times \text{set difficulty} \times \text{set-status factor}$$

$$\text{Relative load} = \frac{\text{effective load}}{\text{current exercise-specific e1RM}}$$

For normal dynamic resistance exercises, these set doses feed the local-muscle allocation and diminishing-return calculations below.

The Resistance Training and Muscular Load paper also defines the principles for maintaining an exercise specific e1RM. To make that baseline operational, working sets of 10 reps or fewer can use the Epley estimate:

$$\text{Set e1RM} = \text{load} \times \left( 1 + \frac{\text{repetitions} + \text{RIR}}{30} \right)$$

RIR (Reps In Reserve - how many more reps you could have done before failure) is included only when the user supplied it and it is 0-3. Without RIR, the usual  $\text{load} \times (1 + \text{repetitions} \div 30)$  estimate should be treated as a lower bound because the set may have stopped well before failure. Multiple repetition tests generally become less accurate as the number of repetitions rises, which is why sets above 10 shouldn't update e1RM. [12]

The exercise baseline should use the highest credible estimate from the previous 90 days, with newer evidence preferred where estimates are similar. It may rise immediately after a clearly stronger performance. If no qualifying evidence appears, it should decay by 0.5% per week after 42 days, with a maximum fall of 10% before asking for new evidence. This slow decay is a design choice: strength can change during time away, although interpreting every light session as lost strength would make relative load jump around for the wrong reason.

Using the exercise involvement coefficients defined in Resistance Training and Muscular Load:

- 0.6 for a small isolation exercise.
- 0.8 for a larger single-joint exercise.
- 1.0 for a multi-joint upper-body exercise.
- 1.2 for a lower-body or full-body compound exercise.

The Resistance Training and Muscular Load paper establishes a difficulty of 1.0 when effort is missing and gives full credit to sets ending around 0-3 RIR. To make the remaining curve implementable, we will use:

- 0-3 RIR: 1.00.
- 4-5 RIR: 0.85.
- 6-7 RIR: 0.70.

- 8 or more RIR: 0.55.

Research finds greater hypertrophy as sets finish closer to failure, while the exact continuous relationship remains unclear and strength outcomes appear less dependent on proximity. RIR estimates also become less accurate further from failure and during higher repetition sets. The broad bands reflect that uncertainty better than pretending each reported RIR is exact. [13] [14]

The set-status factor is normally 1.0. A set explicitly marked as a warm-up receives 0.25. Drop sets remain separate sets at their actual load and repetitions. Assisted exercises subtract assistance from effective load. Dumbbell loads must state whether the number is per hand or total (in the context of an app this should likely be standardised by the app developer for users), unilateral repetitions are recorded per side, and machines remain separate exercise identities. These rules are covered more fully in Resistance Training and Muscular Load because the input semantics have to be correct before any equation is useful.

Session RPE can modify muscular load when set-level RIR is unavailable. The restrained modifier is:

$$\text{Session effort modifier} = \text{clamp}(0.80, 1.20, 0.80 + 0.04 \times \text{session RPE})$$

This would map 5/10 to no change, 10/10 to +20% and 0/10 to -20%. It is applied once after the session's muscular points are summed. If RIR exists for at least 70% of working sets, the session modifier is not applied because difficulty has already been represented at set level. RPE is a valid and useful load-monitoring tool, although results vary by exercise and user familiarity, so keeping its influence within 20% is sensible. [15]

Resistance Training and Muscular Load establishes that set dose should be distributed across the muscles involved. To make that allocation deterministic, assign 70% across primary muscles and 30% across secondary muscles, divided equally within each group unless the exercise library supplies its own specific muscle group shares. An isolation exercise with no meaningful secondary group assigns 100% to its primary muscle. Recent meta-regression work supports giving indirect work partial rather than full credit, although the exact 70/30 split is an implementation value. [16]

For some muscle  $m$ :

$$\text{Raw local dose}(m) = \text{sum of set dose} \times \text{muscle share}(m)$$

Very high local repetition counts should not increase strain in a perfectly straight line. Research on resistance-training volume generally finds greater adaptation with more work, with diminishing returns and differences between strength and hypertrophy outcomes. [16] [17] A simple local transform is therefore useful:

If raw local dose  $\leq 5$  :

$$\text{Local muscular points} = \text{raw local dose}$$

If raw local dose  $> 5$  :

$$\text{Local muscular points} = 5 + 10 \times \ln\left(1 + \frac{\text{raw local dose} - 5}{10}\right)$$

$$\text{Muscular load} = \text{sum of local muscular points across muscles}$$

This curve is linear for the first useful dose and gradually compresses repeated loading of the same tissue. It also means a balanced full-body session can produce more muscular load than repeatedly adding low-value work to one muscle. The threshold of 5 and scale of 10 are initial calibration values. The literature supports diminishing returns but does not provide this exact strain equation, so these numbers should be kept in config and tested against real sessions.

Special forms of resistance work use the routes already defined in Resistance Training and Muscular Load. Bodyweight exercises replace external weight with

$$\text{body mass} \times \text{an exercise-specific effective-load coefficient.}$$

Isometrics use:

$$\text{Timed set dose} = \frac{\text{seconds}}{30} \times \text{relative intensity} \times \text{exercise involvement} \times \text{set difficulty}$$

Loaded carries use

$$\text{carried mass} \times \text{distance}$$

as their external description, then receive both muscular points and independently measured HRR cardio points. Hills, sprints and resisted cardio should only receive muscular load when the activity contains clear resistance or high-force work; ordinary running distance should not be passed through the strength equation.

Generic strength workouts need a fallback because “45 minutes of strength training” does not reveal how much of that time involved lifting. The preferred estimate uses the user’s median muscular points per active minute from detailed sessions in the same category over the previous 90 days:

$$\begin{aligned} \text{Generic muscular load} &= \text{category points per active minute} \\ &\times \text{estimated active minutes} \times \text{session effort modifier} \end{aligned}$$

Use the most recent 8 qualifying sessions where available. Estimated active time should default to 35% of duration for conventional upper, lower or full-body lifting, 60% for calisthenics and 70% for circuits. Personal active time history should replace these fractions once enough timed sets exist. Where the person has no detailed history, starting priors are 0.8 muscular points per active minute for upper body, 1.0 for lower body, 1.1 for full body, 0.9 for circuits and 0.8 for calisthenics. These priors should be labelled low confidence and replaced rather than blended once 3 detailed category sessions exist.

## Putting the channels together

Daily Strain should be an accumulated load score rather than a normal 0-100 quality score. There is no meaningful maximum where 100 means “complete”, and a cap would hide whether someone slightly exceeded their normal load or did twice as much as usual. The simplest combined value is:

$$\text{Daily Strain} = \text{Cardio load} + \text{Muscular load}$$

This addition is a product scale, not a claim that one muscular point and one cardio point are physiologically identical. The muscular formula has been kept in a similar numerical range to the cardio formula, and both channels must always be stored separately. A day can therefore be shown as 64 total strain points, made up of 38 cardio and 26 muscular, rather than losing the composition. The scale should be checked against benchmark sessions before release: an easy walk, a steady run, intervals, heavy lower-body work, upper-body hypertrophy work, a

circuit and a mixed sport session. If one channel consistently dominates sensible comparisons, its global calibration can change without rewriting the underlying dose model.

Cardio and muscular points may occur at the same time. This is intentional rather than double counting because a circuit can load both systems. Double counting happens when several estimates of the same channel are added: HRR plus distance-derived cardio, or detailed set load plus a generic strength prior. The routing hierarchy must allow only one estimate for each time interval and channel.

Confidence should sit beside the load and never change it. Personal Metric Baselines defines the wider confidence framework. For strain, cardio confidence should mainly reflect HR coverage, HRmax source, movement/workout evidence and fallback use. Muscular confidence should reflect whether sets, load, e1RM, exercise identity, muscle allocation and RIR were known. The interface can explain missing inputs, but the underlying point value should remain the estimate produced by the selected route.

Raw strain should also be kept separate from readiness. Poor sleep, low HRV or soreness should not lower the load that was completed. They belong to Readiness Scores and may change the interpretation of that load. The insight layer can say that an otherwise ordinary session may carry a higher recovery cost today, while the recorded strain remains untouched.

The model should provide Daily Strain and a Weekly Strain Target. Weekly targets fit training better because hard, easy and rest days naturally alternate. The target should be based on chronic personal load, with recent weekly load compared against longer history, rather than forcing the same amount every day. Google's maximum of a 28-day rolling mean and an exponentially weighted moving average is a useful approach because one side recovers quickly after a return to training while the other avoids collapsing after a short break. [4] The exact baseline handling, missing-day rules and day-of-week context are already set out in Personal Metric Baselines and should be referenced here rather than repeated.

Apple's 7-day against 28-day comparison supports the same interpretation layer: recent load can be below, steady, above or well above the person's normal amount. [2] These descriptions should not claim an exact injury probability. They explain the size and direction of a change, which is what the underlying evidence can support.

## Conclusion

The strain algorithm should use two separately stored channels and one unbounded total. Cardio load is calculated minute by minute with the modified Banister equation:  $0.64 \times \text{HRR} \times e^{k \times \text{HRR}}$ , using  $k = 1.92$  for males,  $1.67$  for females and  $1.795$  where sex is unavailable. Minutes below 30% HRR receive zero; 30-40% HRR is linearly down-weighted; and incidental minutes require movement evidence. Good HRR data is primary, provider zones or external work replace uncovered intervals, and session-RPE fallback is used only where it describes cardiovascular effort. These estimates are never added together for the same interval.

Detailed dynamic resistance work uses repetitions  $\times$  relative load  $\times$  exercise involvement  $\times$  set difficulty  $\times$  set-status factor. Relative load uses a current exercise-specific e1RM. Involvement begins at 0.6, 0.8, 1.0 or 1.2 by exercise class. Difficulty is 1.0 at 0-3 RIR, 0.85 at 4-5, 0.70 at 6-7 and 0.55 at 8+, while missing effort stays at 1.0 with lower confidence. Warm-ups receive 0.25. Session RPE can modify the result from -20% to +20% only when set-level RIR coverage is poor.

Set dose is allocated 70/30 between primary and secondary muscles, then each local dose

remains linear through 5 points and follows  $5 + 10 \times \ln\left(1 + \frac{\text{dose}-5}{10}\right)$ . Bodyweight, isometric and loaded-carry routes follow Resistance Training and Muscular Load. Generic strength uses the person's recent category-specific muscular points per active minute, with the stated activity fractions and priors until detailed history exists.

The app should calculate Daily Strain = Cardio load + Muscular load, while retaining and displaying the two components. Confidence is reported beside the estimate and does not change it. Readiness changes the interpretation, not the recorded load. Weekly targets and normal-range language use the baseline rules already defined in Personal Metric Baselines.

Some values come directly from established load models and resistance-training research. The 30-40% linear ramp, muscular diminishing-return curve, muscle shares, active-time fractions and generic-session priors are initial implementation choices. They should be stored as versioned configuration and calibrated against benchmark workouts and longitudinal user data rather than becoming invisible constants in the code.

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